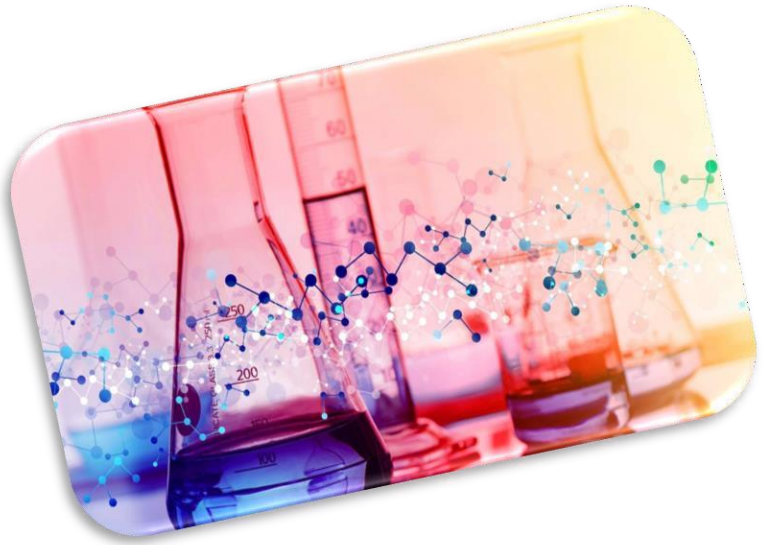
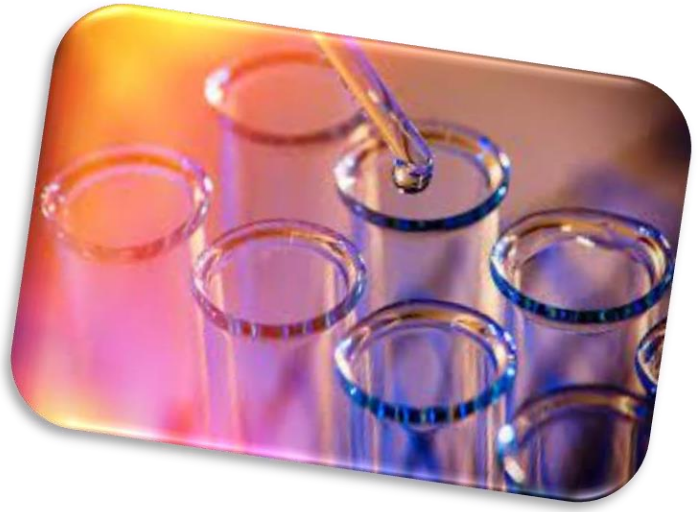
Arab Academy
for Science, Technology & Maritime Transport

Biochemistry
EBA1300





Dr. Mohamed R. Eletmany

PhD in Fiber Science and Polymer Chemistry

NC State Wilson College of Textiles, NC State University, North Carolina, USA

Lecture 4

Cell Transport
(Passive Transport)

Important to know about cell membrane

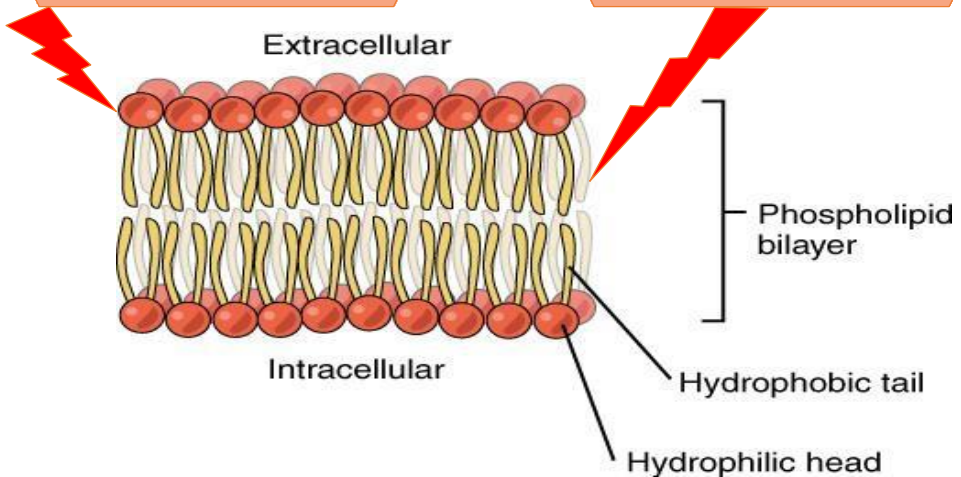
- ✓ Cell membrane is composed of phospholipid bilayer (two layers of phospholipids).
- ✓ The main components of cell membrane are:

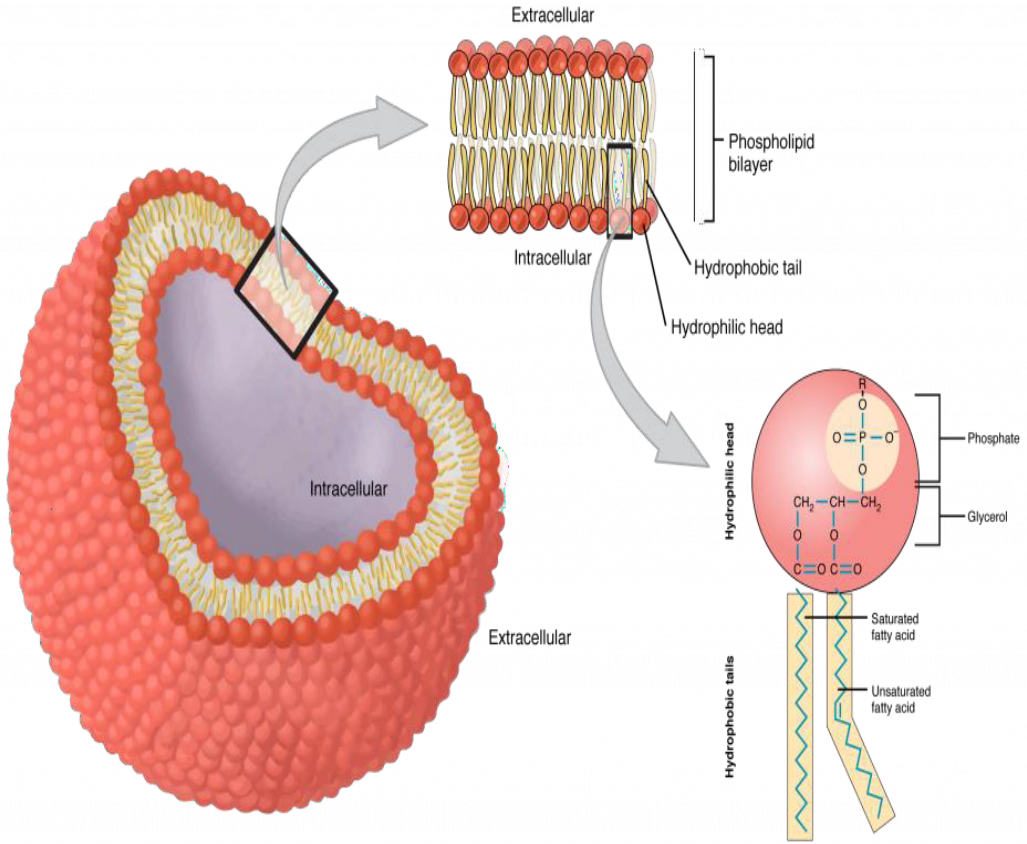
Polar phosphate head (hydrophilic, water loving)

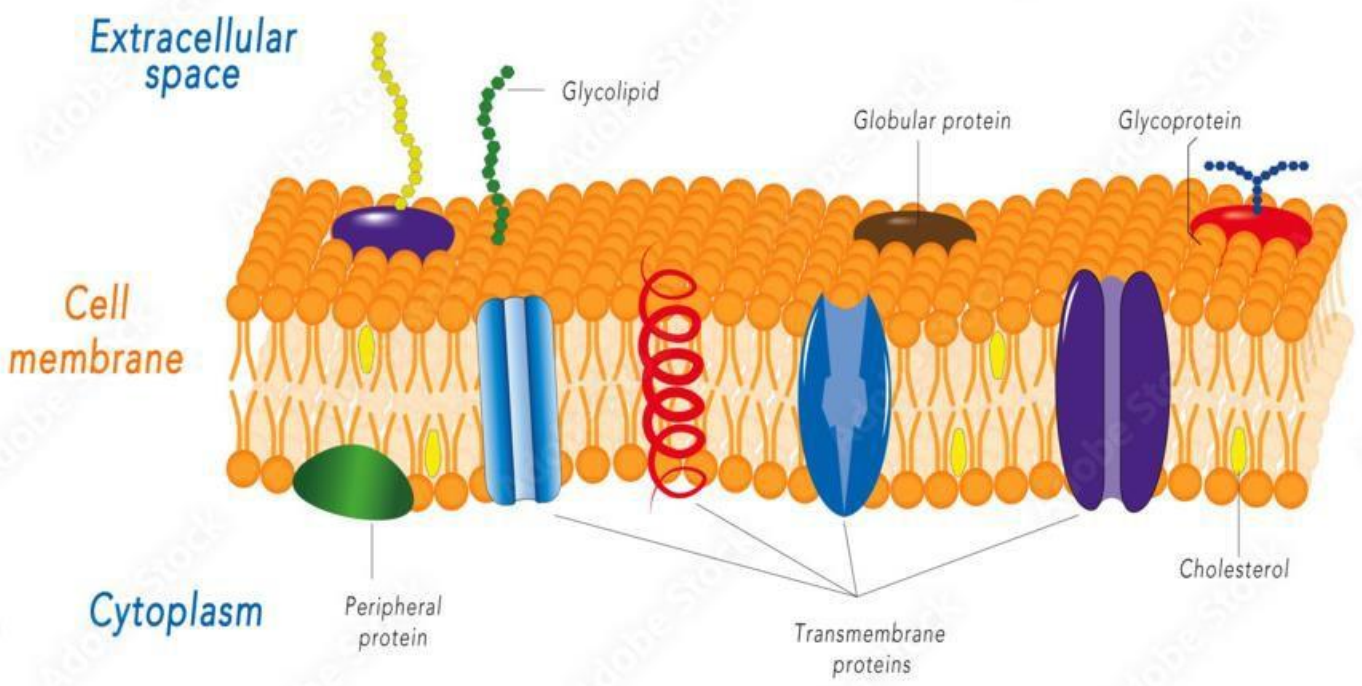
Non-polar Fatty acid tails (hydrophobic or water fearing)

Proteins (may be embedded or peripheral)

Carbohydrates



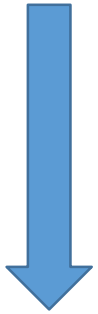




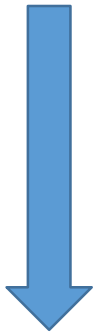
Here, is inside the cell

Important to know about cell membrane

✓ Cell membrane have pores (holes) so,



- It is selective permeable (allows some molecules only to enter or exit the cell)



- Not all molecules can pass through cell membrane, but this depends on size, charge (+ or -) and solubility of the molecule (lipid soluble or not)

What is Cell Transport?

- It is the movement of a substance across the cell membrane.
- The substance can move into the cell or out of the cells.
- Different molecules pass through the plasma membrane or the cell membrane.
- The plasma membrane is highly selective in nature (certain molecules can pass into or out the cell).

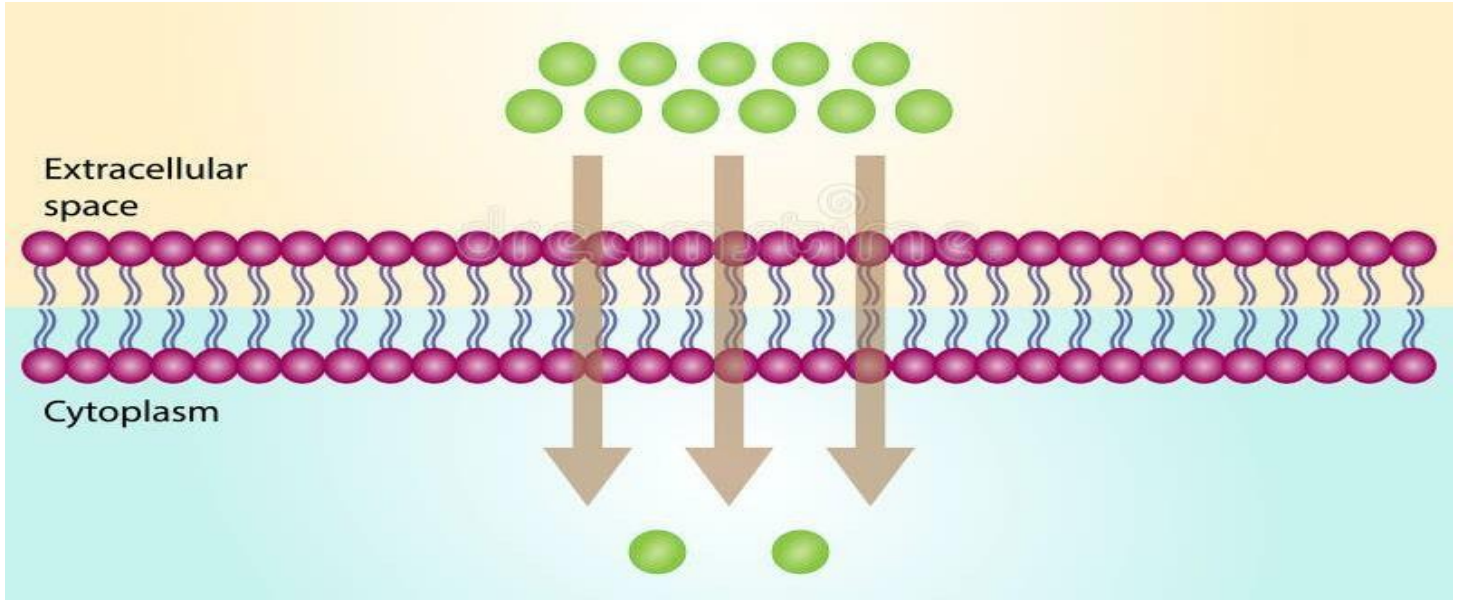
Cell Transport

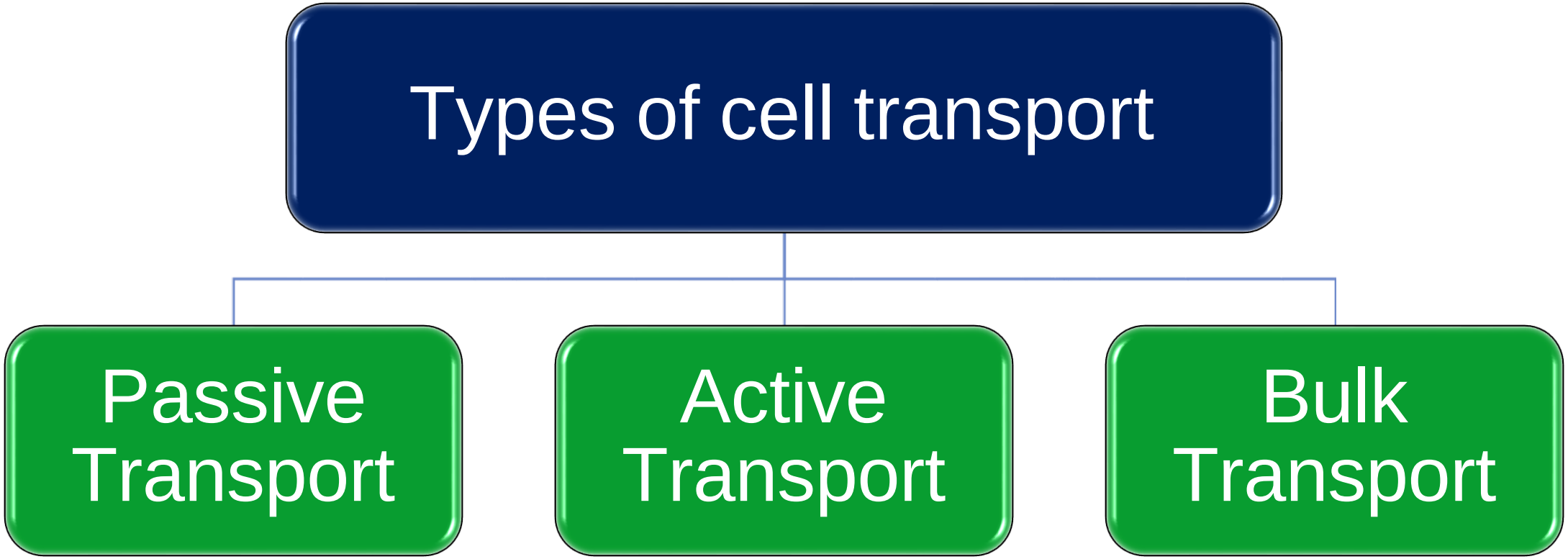
✓ What are molecules that pass easily through the cell ?

Small

Non-charged

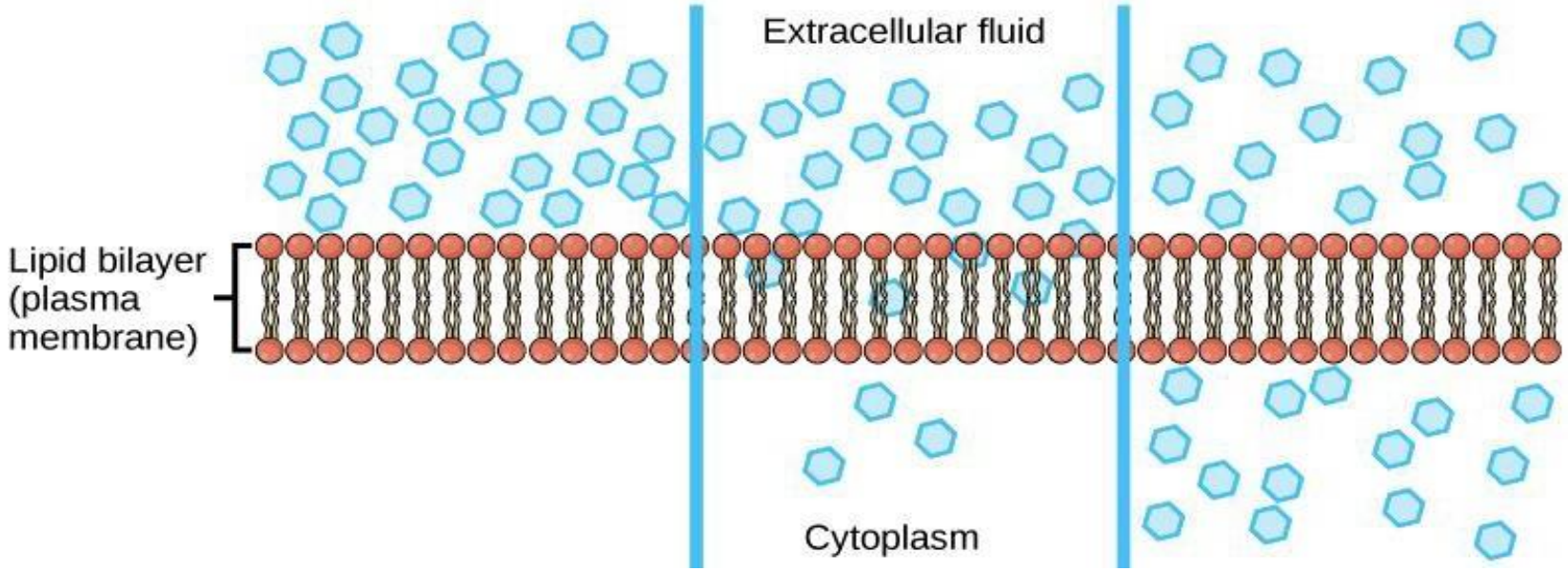
Lipid soluble




Types of cell transport
Passive Transport
Active Transport
Bulk Transport

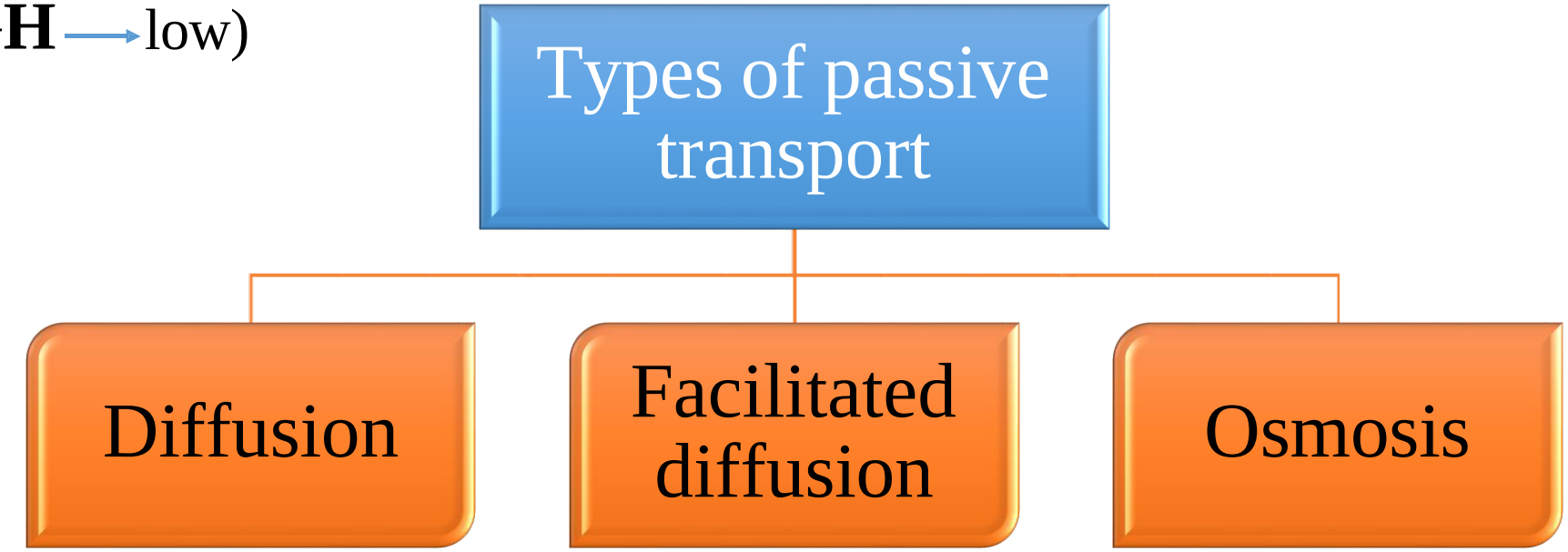
Passive Transport

- In this type of transport, molecules move across the membrane more easily without the need of energy (cell doesn't use energy to transport molecules).
- The movement of substances occurs down their concentration gradient (from area of high concentration of molecules to area of low concentration of molecules), so no need for energy to move.



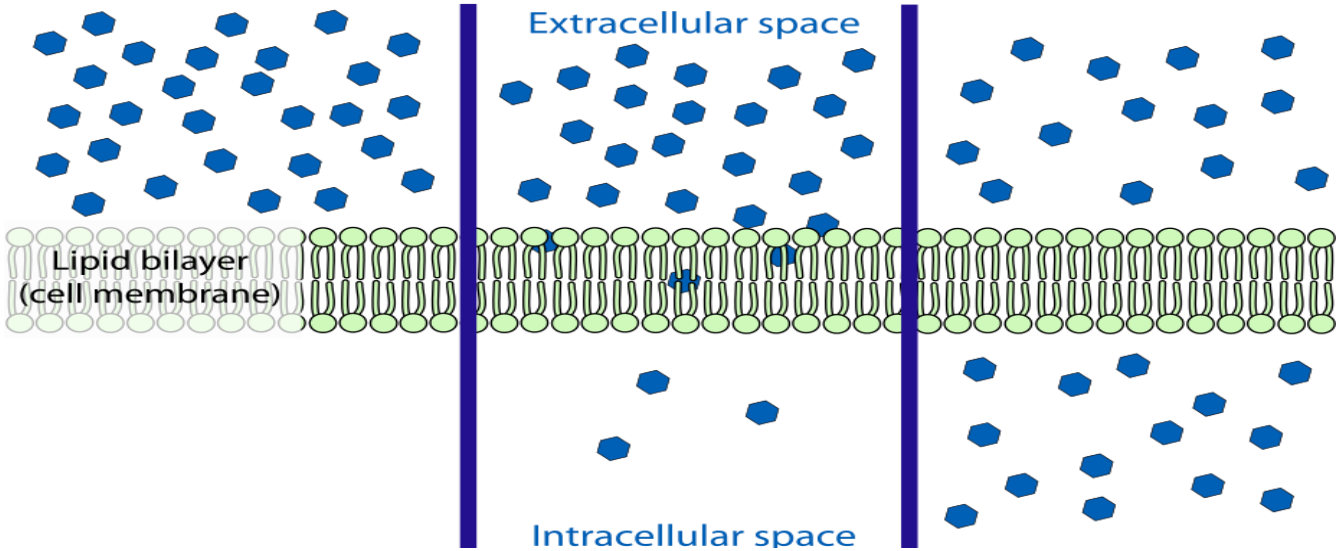
Passive Transport

- No energy needed to transport molecules.
- Molecules move randomly.
- Molecules move from area of High concentration to area of low concentration (down the concentration gradient).
- (**HIGH** → low)



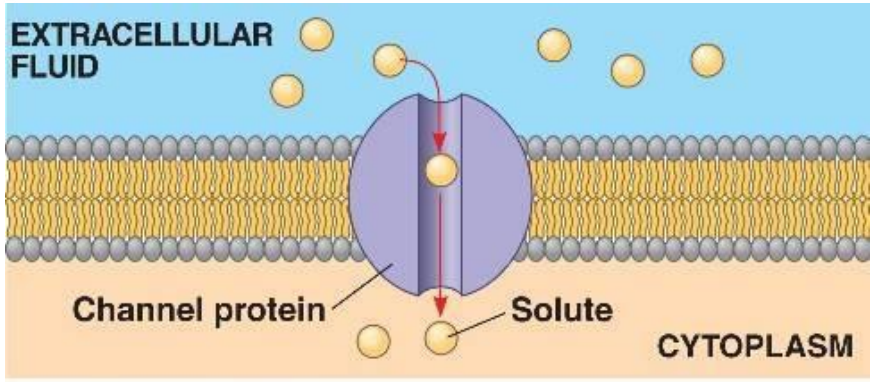
1- Diffusion

- It is the random movement of molecules from the area of high concentration to area of low concentration.
- Small and uncharged molecules such as salts, CO_2, O_2 that are dissolved in H_2O move from their higher to lower concentrations by diffusion.
- The movement of molecules occurs until equilibrium is reached (concentration of molecules outside cell = concentration of molecules inside cell).

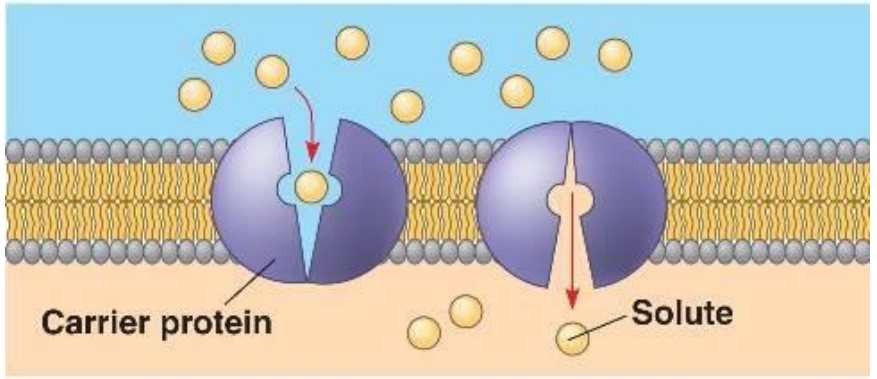


2- Facilitated Diffusion

- This type of diffusion is called facilitated as it needs the help of protein to transport molecules.
- It is important to transport larger molecules such as sugars and amino acids.
- These proteins select molecules that they will pass through the membrane.
- Two types of protein are: carrier proteins and channel proteins.
- No energy needed as the concentration gradient is required.
- Molecules move from higher to lower concentration.



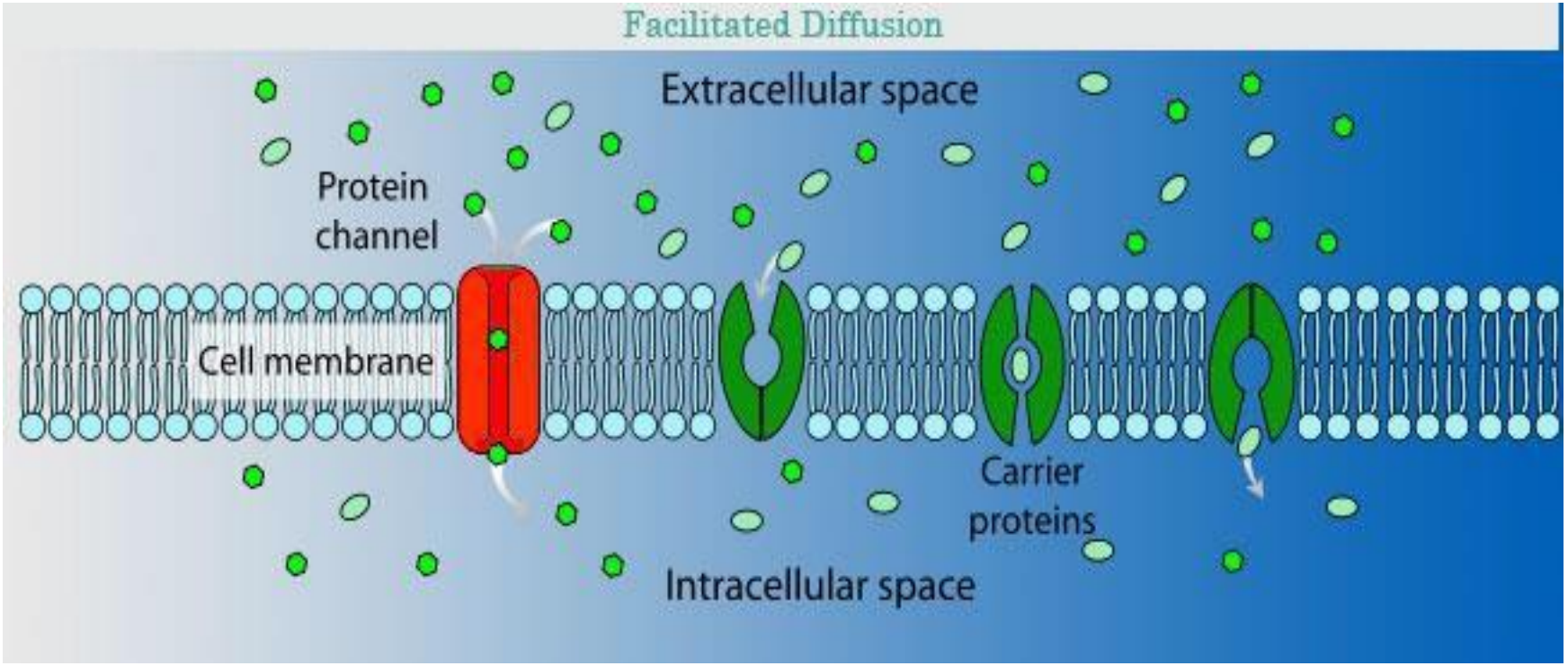
(a) A channel protein



(b) A carrier protein

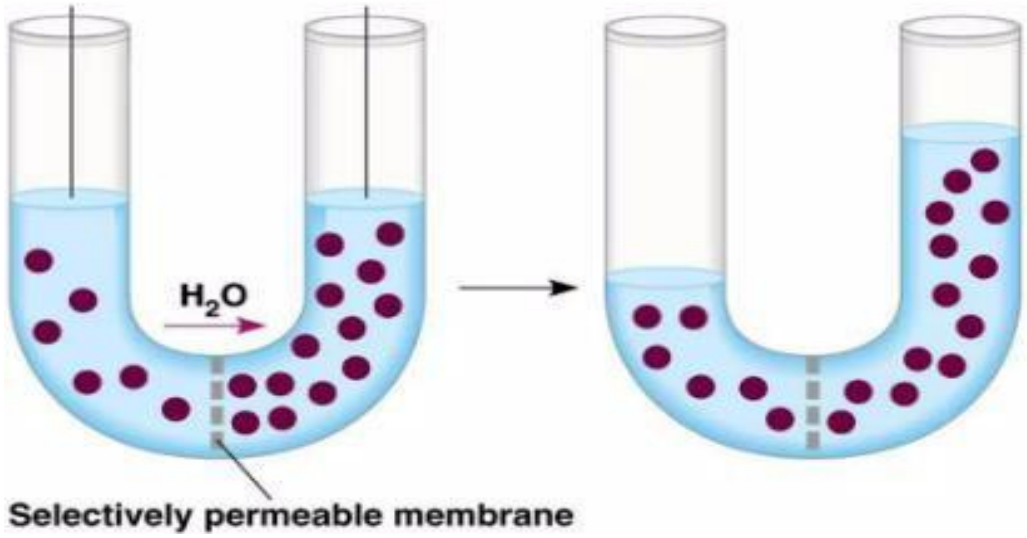
There are two types of proteins in facilitated diffusion

- **Channel proteins:** Contain channels (openings) that pass molecules into the cell.
- **Carrier proteins:** Bind to molecules, and then move them through the membrane into the cell.

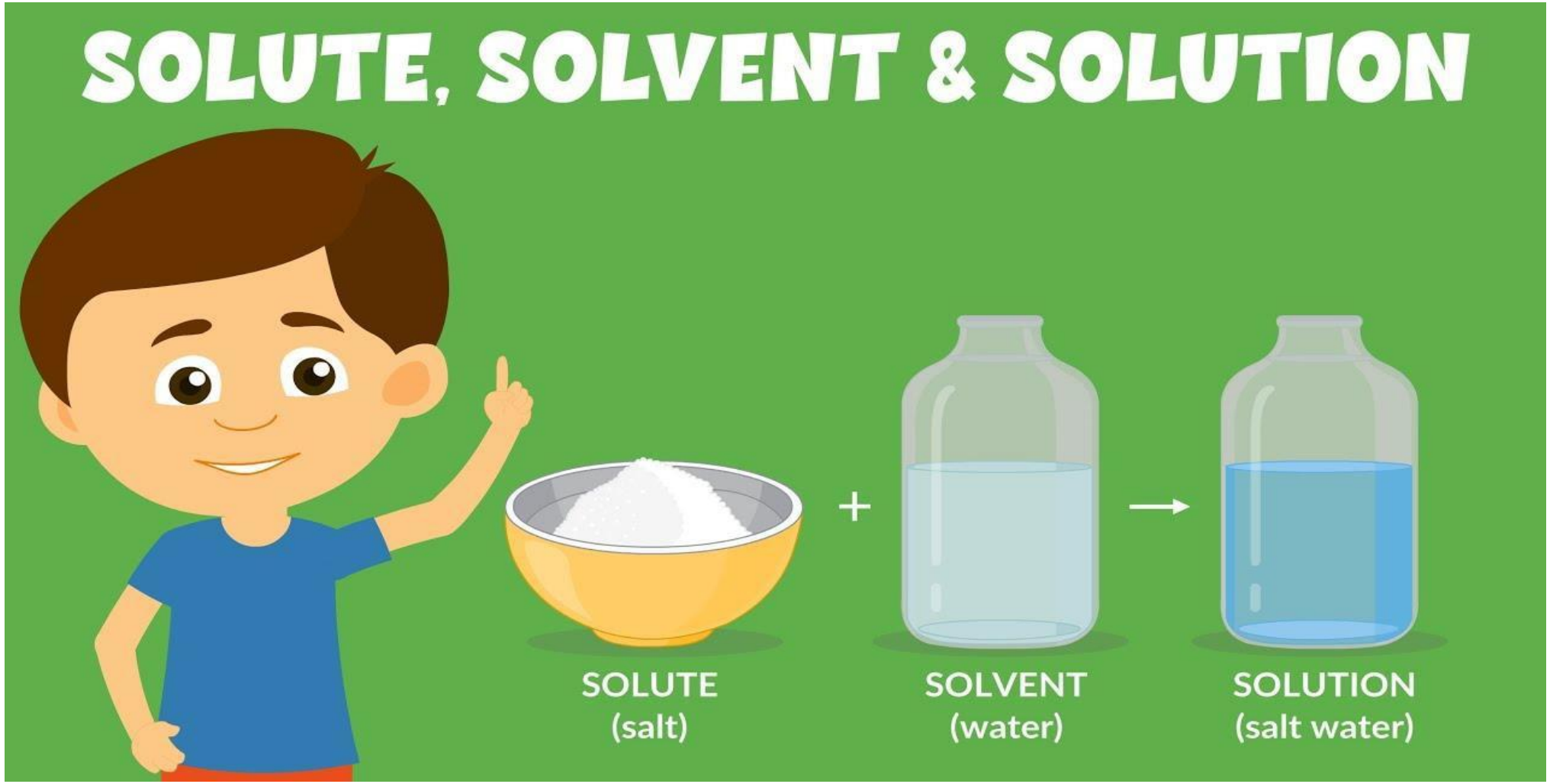


3- Osmosis

- It doesn’t require energy, as it is a passive transport.
- It is the movement of water molecules from the region of high concentration of water to the region of low concentration of water.
- Water moves down its concentration gradient.



Difference between solute, solvent & solution



There are three types of solutions

1 Isotonic solution: The concentration of solute and water is equal inside and outside the cell (equal on both sides).

- Water will move in both sides to keep equilibrium (balance).

There are three types of solutions

2- Hypotonic solution (diluted): Concentration of solute outside the cell is lower than inside the cell, water is more outside the cell.

Outside the cell, water is high and solute is low.

- Water will move into the cell to keep equilibrium, causing cell swelling and cell may explode.

There are three types of solutions

3- Hypertonic solution (concentrated): Concentration of solute outside the cell is higher than inside the cell, water is more inside the cell.

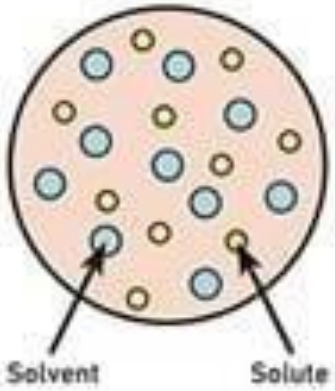
Outside the cell, solute is high and water is low.

- Water will move outside the cell to keep equilibrium, causing cell shrinkage.

Types of Solutions

Isotonic

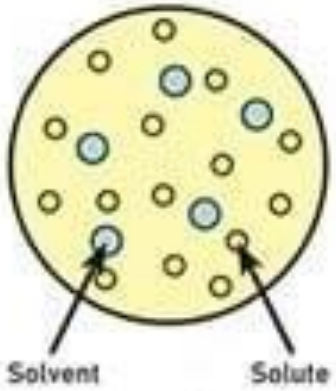
Iso = Same
Tonic = Strength



Same concentration of solutes and solvent

Hypertonic

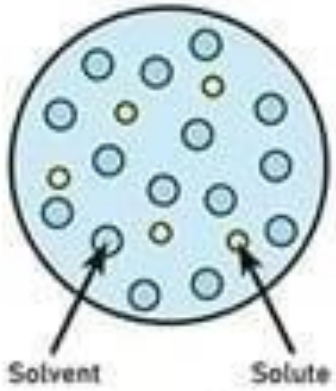
Hyper = More
Tonic = Strength



More solutes and Less solvent

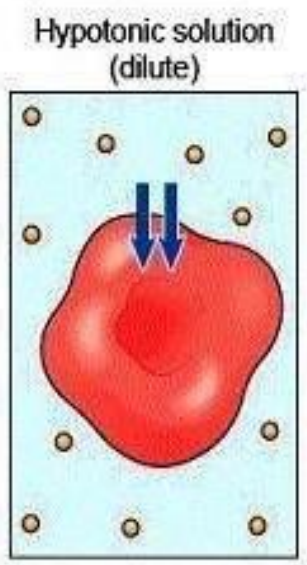
Hypotonic

Hypo = Less
Tonic = Strength

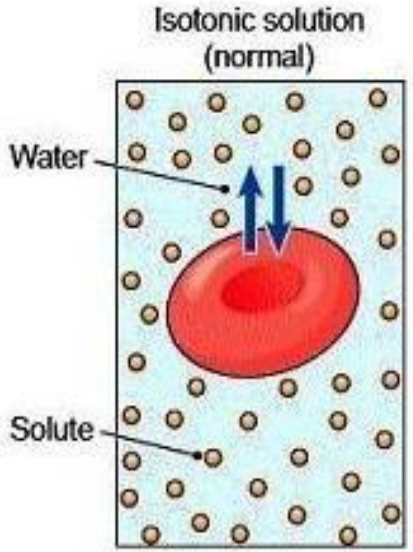


Fewer solutes and More solvent

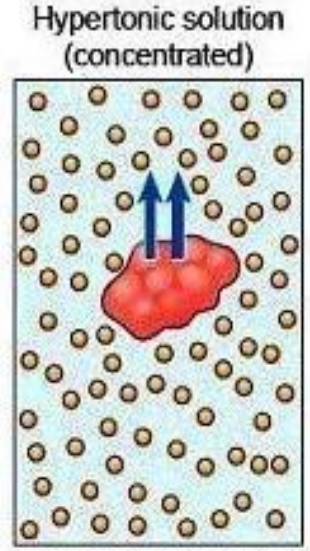
Effect of different solutions on cell



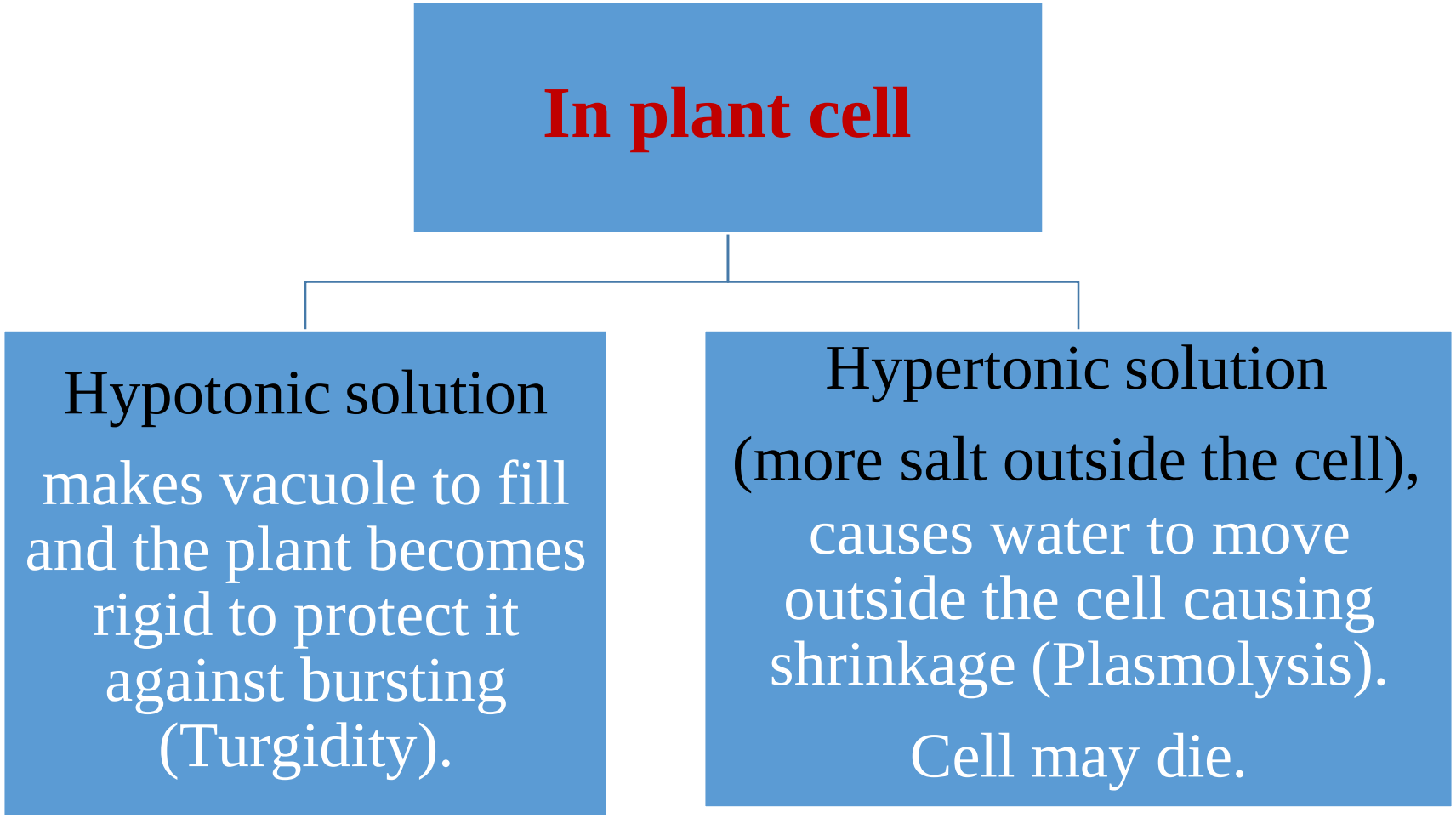
Swollen red blood cell

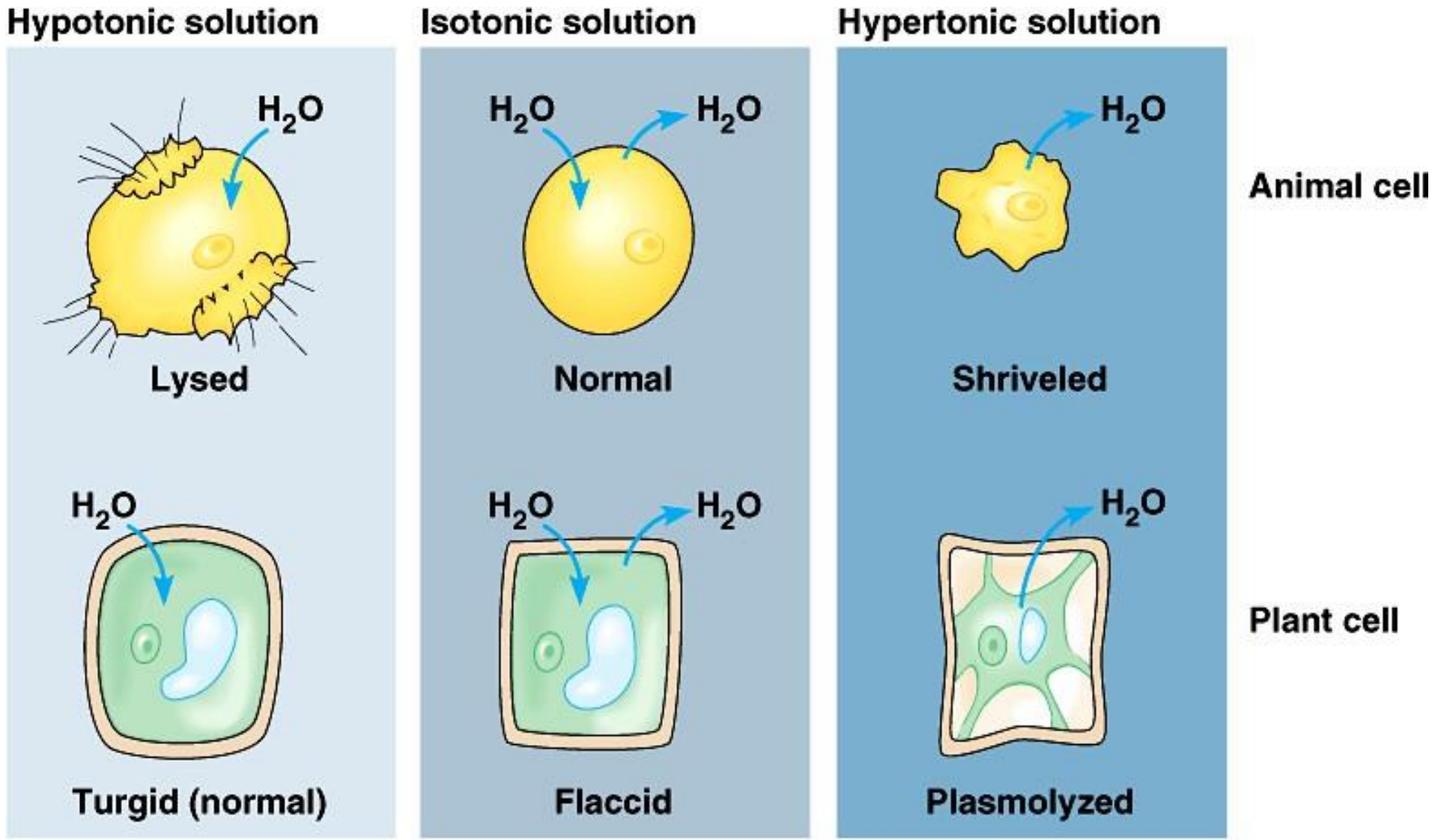


Normal red blood cell



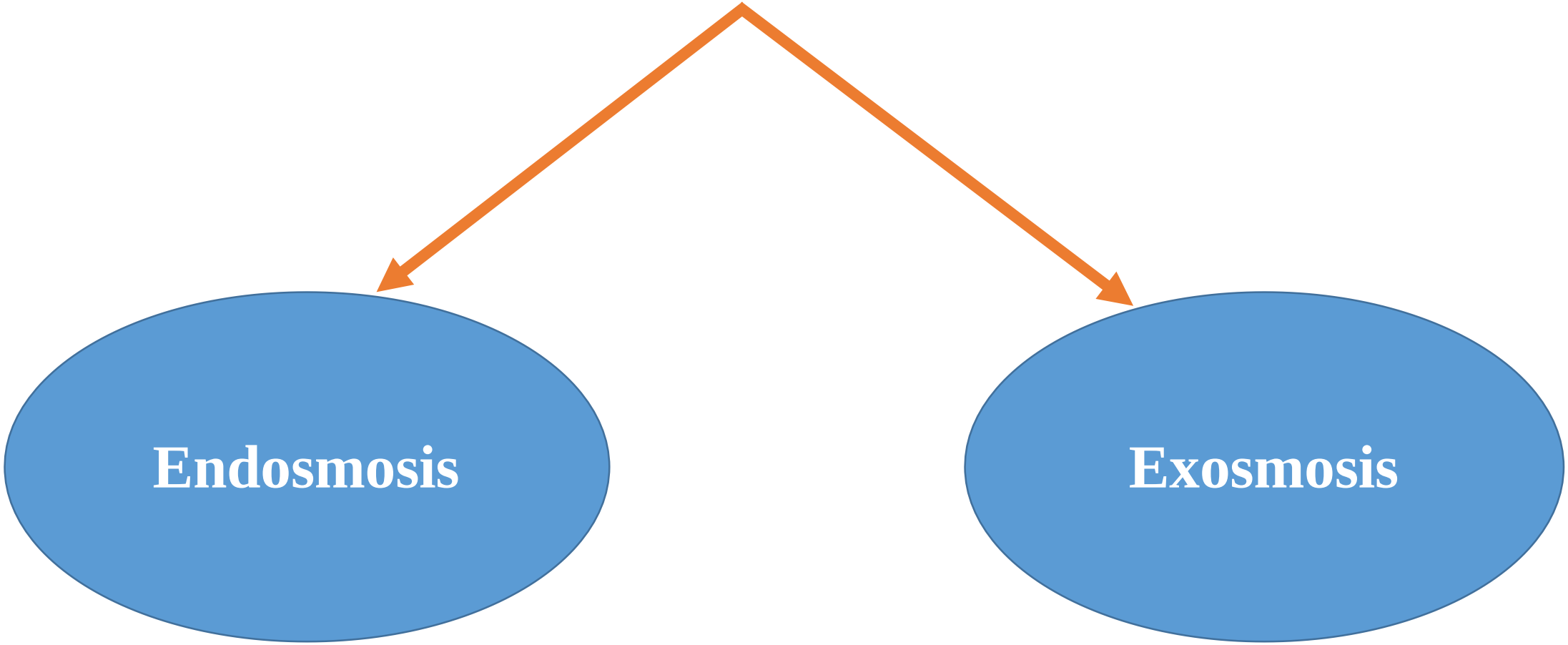
Shrunken (crenated) red blood cell


In plant cell
Hypotonic solution
makes vacuole to fill and the plant becomes rigid to protect it against bursting (Turgidity).
Hypertonic solution
(more salt outside the cell),
causes water to move outside the cell causing shrinkage (Plasmolysis).
Cell may die.



Copyright © Pearson Education, Inc., publishing as Benjamin Cummings.

Two Types of osmosis



Two Types of osmosis

Endosmosis

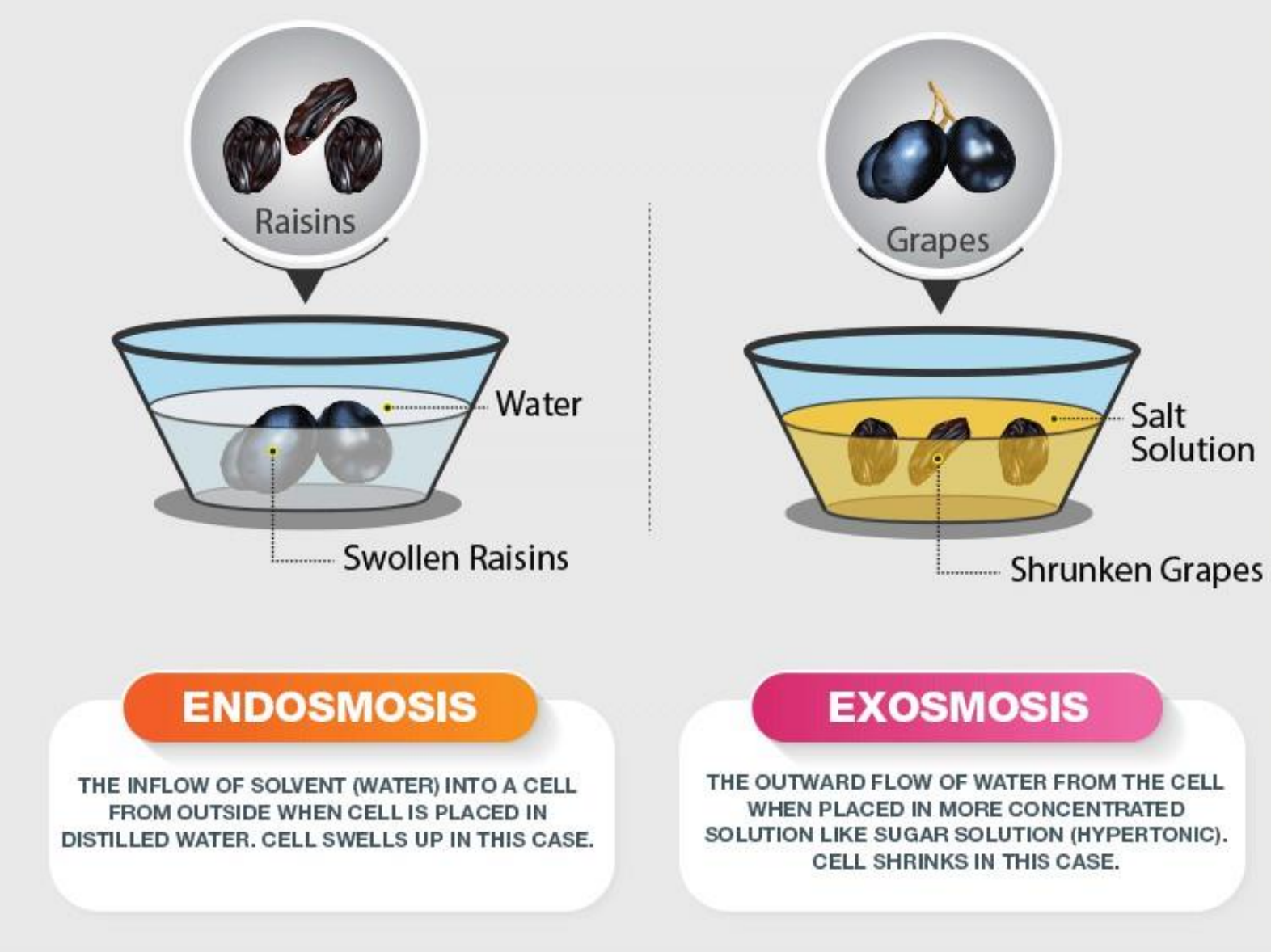
- When a cell is placed in a hypotonic solution, the water moves inside a cell and the cell swells.
- The movement of water inside a cell is known as endosmosis.
- This happens because the solute concentration outside the cell is less than inside cell.

Two Types of osmosis

Exosmosis

- When a cell is placed in a hypertonic solution, the water moves out of the cell and the cell becomes shrunken.
- The movement of water out of the cell is known as exosmosis.
- This happens because the solute concentration of the surrounding solution (outside the cell) is higher than inside the cell.

Endosmosis and Exosmosis



Complete the following:

1 ……..are proteins that bind to molecules, and then move them through the membrane.

2 ……. is composed of lipid bilayer.

3 Animal cells survive only in……solution.

4 ……. , …….. and……molecules pass easily through the cell.

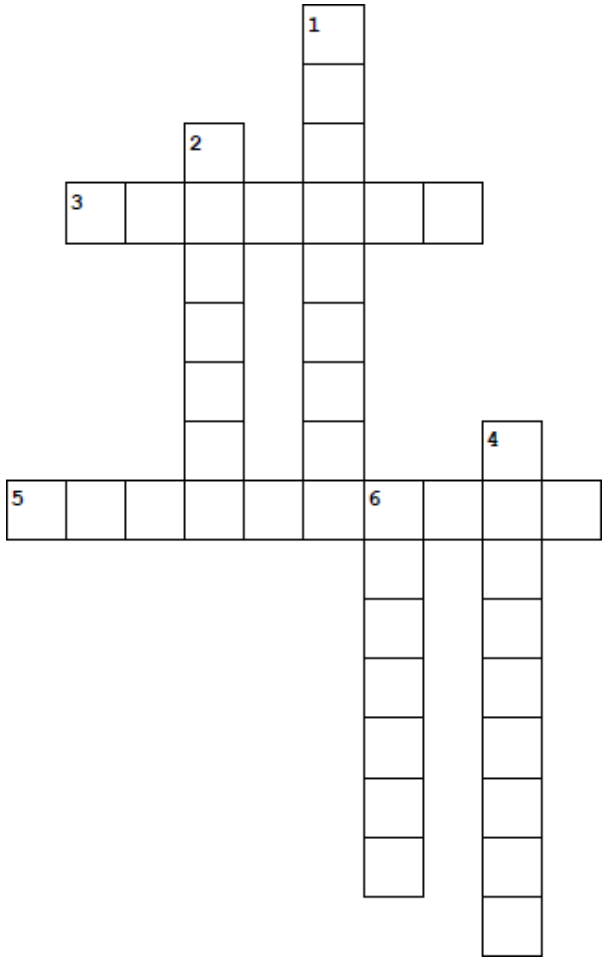
5 When a cell is placed in hypotonic solution, water will move ……. the cell.

6 ……. is the movement of water molecules from the region of high concentration of water to the region of low concentration of water.

7 Types of passive transport are……. , ……..…. and……

8 In passive transport, substances move from ……to……..concentration.

Cross Word



Across

3. proteins Contain channels (openings) that pass molecules into the cell are called

5. In this type of solutions, the concentration of solute outside the cell is higher than inside the cell

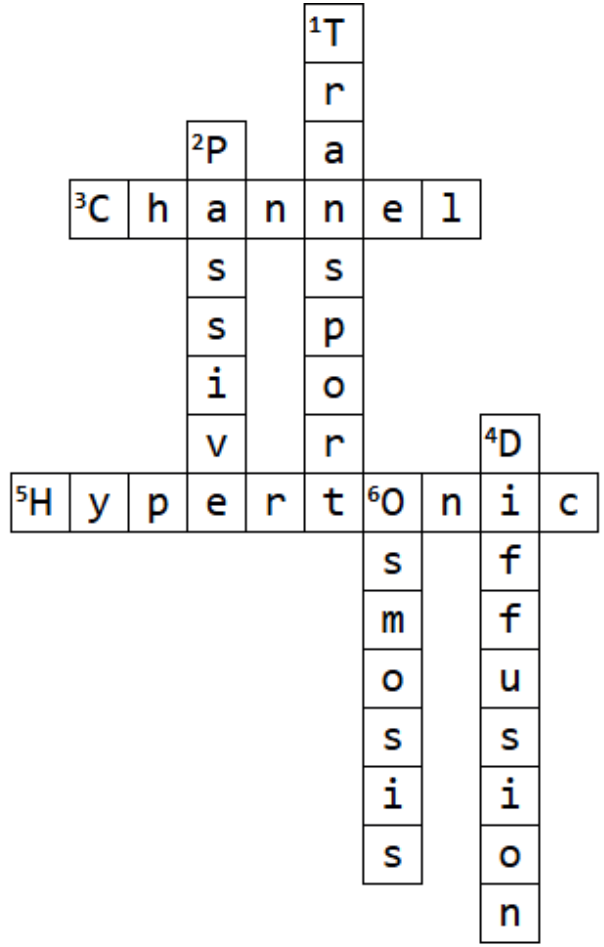
Down

1. The movement of a substance across the cell membrane is called

2. The type of transport where molecules move down their concentration gradient is called

4. Small and uncharged molecules such as salts, CO_2, O_2 that are dissolved in H_2O move by

6. Is the movement of water molecules from the region of high concentration of water to the region of low concentration of water



Across

3. proteins Contain channels (openings) that pass molecules into the cell are called

5. In this type of solutions, the concentration of solute outside the cell is higher than inside the cell

Down

1. The movement of a substance across the cell membrane is called

2. The type of transport where molecules move down their concentration gradient is called

4. Small and uncharged molecules such as salts, CO2, O2 that are dissolved in H2O move by

6. Is the movement of water molecules from the region of high concentration of water to the region of low concentration of water


Good
Luck